Q 1 — **2002**

A superannuation fund pays an interest rate of 8.75% per annum which compounds annually. Stephanie decides to invest \$5000 in the fund at the beginning of each year, commencing on 1 January 2003.

What will be the value of Stephanie's superannuation when she retires on 31 December 2023?

$$A = P\left(1+\frac{r}{100}\right)^n$$
$P = 5000,\ r = 8.75$
$$\therefore A = 5000\left(1+\frac{8.75}{100}\right)^n$$
$$= 5000\,(1.0875)^n$$

$A_1 = 5000\,(1.0875)^{21}$
$A_2 = 5000\,(1.0875)^{20}$
$\vdots$
$A_{21} = 5000\,(1.0875)^1$

Total value on 31 December 2023
$$= 5000\,(1.0875)^{21} + 5000\,(1.0875)^{20} + \ldots + 5000\,(1.0875)^1$$
$$= 5000\,[(1.0875)^{21} + (1.0875)^{20} + \ldots + (1.0875)^1]$$
$$= 5000\,[(1.0875)^1 + (1.0875)^2 + \ldots + (1.0875)^{21}]$$
$$= 5000\left[\frac{a(r^n-1)}{r-1}\right]$$
where $a = 1.0875,\ r = 1.0875,\ n = 21$
$$= 5000\left[\frac{1.0875(1.0875^{21}-1)}{1.0875-1}\right]$$
$= \$299\,605$ to the nearest dollar. (by calc.)

Q 2 — **2004**

Betty decides to set up a trust fund for her grandson, Luis. She invests \$80 at the beginning of each month. The money is invested at 6% per annum, compounded monthly.

The trust fund matures at the end of the month of her final investment, 25 years after her first investment. This means that Betty makes 300 monthly investments.

(i) After 25 years, what will be the value of the first \$80 invested?

(ii) By writing a geometric series for the value of all Betty's investments, calculate the final value of Luis' trust fund.

(i) $A = P(1+r)^n$.
$P = 80$, $r = 6\%$ per annum
$= \frac{6\%}{12}$ per month
$= 0{\cdot}5\%$
$= 0{\cdot}005$,
$n = 25$ years
$= 300$ months.

$A = 80(1+0{\cdot}005)^{300}$
$= 80(1{\cdot}005)^{300}$
$= 357{\cdot}197\,585\ldots$
$= \$357{\cdot}20$ (to the nearest cent).

$\therefore$ After 25 years, the value of the first \$80 invested will be \$357·20.

(ii) Let A_n be the value of the nth instalment at the end of the term.

1st \$80 invested: $A_1 = 80(1{\cdot}005)^{300}$
2nd \$80 invested: $A_2 = 80(1{\cdot}005)^{299}$
3rd \$80 invested: $A_3 = 80(1{\cdot}005)^{298}$
$\vdots$
last \$80 invested: $A_{300} = 80(1{\cdot}005)^1$.

Total $= A_{300} + A_{299} + A_{298} + \cdots + A_1$
$= 80(1{\cdot}005) + 80(1{\cdot}005)^2 + 80(1{\cdot}005)^3 + \cdots + 80(1{\cdot}005)^{300}$
$= 80(1{\cdot}005 + 1{\cdot}005^2 + 1{\cdot}005^3 + \cdots + 1{\cdot}005^{300})$
$= 80 \times S_n$, S_n is a geometric series, with $a = 1{\cdot}005$, $r = 1{\cdot}005$, $n = 300$.
$$= 80 \times \frac{a(r^n-1)}{r-1}$$
$$= 80 \times \frac{1{\cdot}005(1{\cdot}005^{300}-1)}{1{\cdot}005-1}$$
$= 55\,716{\cdot}714\,58\ldots$
$= \$55\,716{\cdot}71$ (to the nearest cent).

$\therefore$ Final value of the trust fund will be \$55 716·71.

Q 3 2007

Mr and Mrs Caine each decide to invest some money each year to help pay for their son's university education. The parents choose different investment strategies.

(i) Mr Caine makes 18 yearly contributions of $1000 into an investment fund. He makes his first contribution on the day his son is born, and his final contribution on his son's seventeenth birthday. His investment earns 6% compound interest per annum.

Find the total value of Mr Caine's investment on his son's eighteenth birthday.

(ii) Mrs Caine makes her contributions into another fund. She contributes $1000 on the day of her son's birth, and increases her annual contribution by 6% each year. Her investment also earns 6% compound interest per annum.

Find the total value of Mrs Caine's investment on her son's third birthday (just before she makes her fourth contribution).

(iii) Mrs Caine also makes her final contribution on her son's seventeenth birthday.

Find the total value of Mrs Caine's investment on her son's eighteenth birthday.

$$(c)(i)\ A=1000(1.06)^{18}+1000(1.06)^{17}+\ldots+1000(1.06)$$
$$=1000(1.06)\left[1+1.06+\ldots+1.06^{17}\right]$$
$$=1000(1.06)\ \frac{1\left(1.06^{18}-1\right)}{1.06-1}$$
$$=\$32759.99$$

$$(ii)\ A_3=1000(1.06)^3+\left[1000(1.06)\right]\ (1.06)^2$$
$$\left[1000(1.06)^2\right]\ (1.06)^1$$
$$=1000(1.06)^3\quad 1000(1.06)^3\quad 1000(1.06)^3$$
$$=3\quad 1000(1.06)^3$$
$$=\$3573.05$$

$$(iii)\ A_{18}=18\quad 1000(1.06)^{18}$$
$$=\$51378.10$$

Q 4 **2011**

When Jules started working she began paying \$100 at the beginning of each month into a superannuation fund.

The contributions are compounded monthly at an interest rate of 6% per annum.

She intends to retire after having worked for 35 years.

(i) Let \$$P$ be the final value of Jules's superannuation when she retires after 35 years (420 months).

Show that \$$P$ = \$143 183 to the nearest dollar.

(ii) Fifteen years after she started working Jules read a magazine article about retirement, and realised that she would need \$800 000 in her fund when she retires. At the time of reading the magazine article she had \$29 227 in her fund. For the remaining 20 years she intends to work, she decides to pay a total of \$$M$ into her fund at the beginning of each month. The contributions continue to attract the same interest rate of 6% per annum, compounded monthly.

At the end of n months after starting the new contributions, the amount in the fund is \$$A_n$.

(1) Show that $A_2 = 29\,227 \times 1.005^2 + M(1.005 + 1.005^2)$.

(2) Find the value of M so that Jules will have \$800 000 in her fund after the remaining 20 years (240 months).

Q8ci $R = 1 + \dfrac{6}{12\times 100} = 1.005$

End of nth month	\$$P_n$
1	100×1.005^1
2	$100\times1.005^1+100\times1.005^2$
3	$100\times1.005^1+100\times1.005^2+100\times1.005^3$
…	…
420	$\dfrac{100\times1.005\left(1.005^{420}-1\right)}{0.005}\approx 143183$

Q8cii1

End of nth month	\$$A_n$
1	$29227\times1.005+M\times1.005$
2	$(29227\times1.005+M+M\times1.005)\times1.005$ $=29227\times1.005^2+M\left(1.005+1.005^2\right)$

Q8cii2 At the end of 240[th] month after starting the new contributions, A_{240}

$$=29227\times1.005^{240}+M\left(1.005+1.005^2+\ldots+1.005^{240}\right)=800000$$

$$\therefore\ M=\frac{800000-29227\times1.005^{240}}{1.005+1.005^2+\ldots+1.005^{240}}=\frac{703252.6538}{464.3511}=1514.48$$

Q 1 — 2001

Helen sets up a prize fund with a single investment of \$1000 to provide her school with an annual prize valued at \$72. The fund accrues interest at a rate of 6% per annum, compounded annually. The first prize is awarded one year after the investment is set up.

(i) Calculate the balance in the fund at the beginning of the second year.

(ii) Let $\$B_n$ be the balance in the fund at the end of n years (and after the nth prize has been awarded). Show that $B_n = 1200 - 200 \times (1.06)^n$.

(iii) At the end of the tenth year (and after the tenth prize has been awarded) it is decided to increase the prize value to \$90. For how many more years can the prize fund be used to award the prize?

(a) (i) Balance at the beginning of the second year

$$= 1000 + 6\% \text{ of } 1000 - 72$$
$$= 1000 \times 1.06 - 72$$
$$= \$988.$$

(ii)
$$B_1 = 1000 \times 1.06 - 72.$$
$$B_2 = (1000 \times 1.06 - 72) \times 1.06 - 72$$
$$= 1000(1.06)^2 - 72(1.06) - 72$$
$$= 1000(1.06)^2 - 72(1.06 + 1).$$
$$B_3 = [1000(1.06)^2 - 72(1.06 + 1)] \times 1.06 - 72$$
$$= 1000(1.06)^3 - 72(1.06^2 + 1.06) - 72$$
$$= 1000(1.06)^3 - 72(1.06^2 + 1.06 + 1).$$
$$\therefore B_n = 1000(1.06)^n - 72(1.06^{n-1} + 1.06^{n-2} + \cdots + 1.06^2 + 1.06 + 1)$$
$$= 1000(1.06)^n - 72(1 + 1.06 + 1.06^2 + \cdots + 1.06^{n-1})$$
$$= 1000(1.06)^n - 72 \times \frac{a(r^n - 1)}{r - 1}, \quad [a = 1,\ r = 1.06],$$
$$= 1000(1.06)^n - 72 \times 1 \times \frac{(1.06^n - 1)}{1.06 - 1}$$
$$= 1000(1.06)^n - 72\left[\frac{1.06^n - 1}{0.06}\right]$$
$$= 1000(1.06)^n - 1200(1.06^n - 1).$$
$$\therefore B_n = 1200 - 200(1.06)^n.$$

(iii) Balance at the end of the 10th year:

$$B_{10} = 1200 - 200(1.06)^{10}$$
$$= \$841.83.$$

At the end of 1 more year:

$$B_{11} = 841.83(1.06) - 90$$

and
$$B_{12} = [841.83(1.06) - 90] \times 1.06 - 90$$
$$= 841.83(1.06)^2 - 90(1.06 + 1).$$
$$B_{10+n} = 841.83(1.06)^n - 90(1.06^{n-1} + 1.06^{n-2} + \cdots + 1.06 + 1)$$
$$= 841.83(1.06)^n - 90(1 + 1.06 + \cdots + 1.06^{n-1})$$
$$= 841.83(1.06)^n - 90 \times 1 \times \left(\frac{1.06^n - 1}{1.06 - 1}\right)$$
$$= 841.83(1.06)^n - 90\left(\frac{1.06^n - 1}{0.06}\right)$$
$$= 841.83(1.06)^n - 1500(1.06^n - 1).$$
$$\therefore B_{10+n} = 1500 - 658.17(1.06)^n.$$

To find n, let $B_{10+n} = 0$.

$$0 = 1500 - 658.17(1.06)^n$$
$$(1.06)^n = \frac{1500}{658.17}$$
$$\ln(1.06)^n = \ln\left(\frac{1500}{658.17}\right)$$
$$n\ln(1.06) = \ln\left(\frac{1500}{658.17}\right)$$
$$n = \ln\left(\frac{1500}{658.17}\right) \div \ln 1.06$$
$$= 14.$$

$\therefore$ The prize can be awarded for 14 more years.

Q 2 — 2004

Weelabarrabak Shire Council borrowed \$3 000 000 at the beginning of 2005. The annual interest rate is 12%. Each year, interest is calculated on the balance at the beginning of the year and added to the balance owing. The debt is to be repaid by equal annual repayments of \$480 000, with the first repayment being made at the end of 2005.

Let A_n be the balance owing after the n-th repayment.

(i) Show that $A_2 = (3 \times 10^6)(1.12)^2 - (4.8 \times 10^5)(1 + 1.12)$.

(ii) Show that $A_n = 10^6\left[4 - (1.12)^n\right]$.

(iii) In which year will Weelabarrabak Shire Council make the final repayment?

(i)
$$A_1 = (3 \times 10^6)\left(1 + \frac{12}{100}\right) - 480\,000$$
$$= (3 \times 10^6)(1.12) - 4.8 \times 10^5.$$
$$A_2 = A_1(1.12) - 4.8 \times 10^5$$
$$= [(3 \times 10^6)(1.12) - 4.8 \times 10^5] \times 1.12 - 4.8 \times 10^5$$
$$= (3 \times 10^6)(1.12)^2 - 1.12 \times 4.8 \times 10^5 - 4.8 \times 10^5$$
$$= (3 \times 10^6)(1.12)^2 - (4.8 \times 10^5)(1 + 1.12).$$

(ii)
$$A_3 = A_2(1.12) - 4.8 \times 10^5$$
$$= (3 \times 10^6)(1.12)^3 - (4.8 \times 10^5) \times (1 + 1.12 + 1.12^2).$$
$$\therefore A_n = (3 \times 10^6)(1.12)^n - (4.8 \times 10^5) \times (1 + 1.12 + 1.12^2 + \cdots + 1.12^{n-1}).$$

Now $1 + 1.12 + 1.12^2 + \cdots + 1.12^{n-1}$ is a geometric series with $a = 1$, $r = 1.12$ and n terms.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$
$$= \frac{1(1.12^n - 1)}{1.12 - 1}$$
$$\therefore S_n = \frac{(1.12^n - 1)}{0.12}.$$

Substituting S_n into the expression for A_n:

$$A_n = (3 \times 10^6)(1.12)^n - (4.8 \times 10^5) \times \frac{(1.12^n - 1)}{0.12}$$
$$= (3 \times 10^6)(1.12)^n - (4 \times 10^6)(1.12^n - 1)$$
$$= 10^6[3(1.12)^n - 4(1.12)^n + 4]$$
$$= 10^6[4 - (1.12)^n].$$

(iii) The loan is repaid when $A_n = 0$.

$$10^6[4 - (1.12)^n] = 0$$
$$4 - (1.12)^n = 0$$
$$(1.12)^n = 4$$

ie.
$$\ln(1.12)^n = \ln 4$$
$$n\ln(1.12) = \ln 4$$
$$n = \frac{\ln 4}{\ln(1.12)}$$
$$n = 12.232\,510\,75\ldots$$

$\therefore$ The final payment is made during the 13th year which is during 2017 (since the first repayment is made at the end of 2005).

Q 3 **2003**

Barbara borrows \$120 000 to be repaid over a period of 25 years at 6% per annum reducible interest. Each year there are k regular repayments of \$$F$. Interest is calculated and charged just before each repayment.

(i) Write down an expression for the amount owing after two repayments.

(ii) Show that the amount owing after n repayments is

$$A_n = 120\,000\,\alpha^n - \frac{kF(\alpha^n - 1)}{0.06}, \quad \text{where } \alpha = 1 + \frac{0.06}{k}.$$

(iii) Calculate the amount of each repayment if the repayments are made quarterly (ie. $k = 4$).

(iv) How much would Barbara have saved over the term of the loan if she had chosen to make monthly rather than quarterly repayments?

(a) (i) $A_1 = 120\,000\left(1+\frac{0.06}{k}\right)^1 - F$

$$A_2 = \left[120\,000\left(1+\frac{0.06}{k}\right)^1 - F\right]\left(1+\frac{0.06}{k}\right) - F$$

$$= 120\,000\left(1+\frac{0.06}{k}\right)^2 - \left(1+\frac{0.06}{k}\right)F - F$$

(ii) $A_1 = 120\,000\,\alpha - F$

$A_2 = 120\,000\,\alpha^2 - \alpha F - F$

$A_3 = (120\,000\,\alpha^2 - \alpha F - F)(\alpha) - F$

$= 120\,000\,\alpha^3 - \alpha^2 F - \alpha F - F$

$$\therefore A_n = 120\,000\alpha^n - \alpha^{n-1}F - \alpha^{n-2}F - \ldots - \alpha F - F$$

$$= 120\,000\alpha^n - F(\alpha^{n-1} + \alpha^{n-2} + \ldots + \alpha + 1)$$

$$= 120\,000\alpha^n - F(1 + \alpha + \alpha^2 + \ldots + \alpha^{n-1})$$

$$= 120\,000\alpha^n - F\left(\frac{a(r^n-1)}{r-1}\right)$$

where $a = 1$, $r = \alpha$ and $n = n$

$$= 120\,000\alpha^n - F\left(\frac{1(\alpha^n-1)}{\alpha-1}\right)$$

$$= 120\,000\alpha^n - F\left(\frac{\alpha^n-1}{1+\frac{0.06}{k}-1}\right)$$

$$= 120\,000\alpha^n - F\left(\frac{\alpha^n-1}{\frac{0.06}{k}}\right)$$

$$= 120\,000\alpha^n - \frac{kF(\alpha^n-1)}{0.06}.$$

(iii) $n = 25 \times 4,\ k = 4,\ A_n = 0$

$= 100$

$\therefore \alpha = 1 + \frac{0.06}{4}$

$= 1.015$

$$\therefore 0 = 120\,000(1.015)^{100} - \frac{4F(1.015^{100}-1)}{0.06}$$

$$= 7200(1.015)^{100} - 4F(1.015^{100}-1)$$

$$4F(1.015^{100}-1) = 7200(1.015)^{100}$$

$$F = \frac{7200\,(1.015)^{100}}{4\,(1.015^{100}-1)}$$

$= \$2324.47.$ (by calc.)

(iv) $n = 25 \times 12,\ k = 12,\ A_n = 0$

$= 300$

$\therefore \alpha = 1 + \frac{0.06}{12}$

$= 1.005$

$$\therefore 0 = 120\,000(1.005)^{300} - \frac{12F(1.005^{300}-1)}{0.06}$$

$$= 7200(1.005)^{300} - 12F(1.005^{300}-1)$$

$$12F(1.005^{300}-1) = 7200(1.005)^{300}$$

$$\therefore F = \frac{7200\,(1.005)^{300}}{12\,(1.005^{300}-1)}$$

$= \$773.16$ (by calc.)

Amount repaid if quarterly repayments:
$= \$2324.47 \times 100$
$= \$232\,447$

Amount repaid if monthly repayments:
$= \$773.16 \times 300$
$= \$231\,948$

Saving $= \$232\,447 - \$231\,948$
$= \$499.$

Q 4 **2006**

Joe borrows \$200 000 which is to be repaid in equal monthly instalments. The interest rate is 7.2% per annum reducible, calculated monthly.

It can be shown that the amount, $\$A_n$, owing after the nth repayment is given by the formula:

$$A_n = 200\,000r^n - M(1 + r + r^2 + \cdots + r^{n-1}),$$

where $r = 1.006$ and $\$M$ is the monthly repayment. (Do NOT show this.)

(i) The minimum monthly repayment is the amount required to repay the loan in 300 instalments.
Find the minimum monthly repayment.

(ii) Joe decides to make repayments of \$2800 each month from the start of the loan.
How many months will it take for Joe to repay the loan?

(b) (i)

$$A_n = 200\,000r^n - M\left(1+r+r^2+\ldots+r^{n-1}\right)$$

$$0 = 200\,000(1.006)^{300} - M\left(1+1.006+1.006^2+..1.006^{299}\right)$$

$$0 = 200\,000(1.006)^{300} - M\left(\frac{1.006^{300}-1}{1.006-1}\right)$$

$$M\left(\frac{1.006^{300}-1}{0.006}\right) = 200\,000(1.006)^{300}$$

$$M = \frac{200\,000(1.006)^{300}\,0.006}{1.006^{300}-1}$$

$$= \$1439.18$$

(ii)

$$A_n = 0 = 200\,000(1.006)^n - 2800\left(\frac{1.006^n-1}{0.006}\right)$$

$$0 = 1200(1.006)^n - 2800(1.006)^n + 2800$$

$$1600(1.006)^n = 2800$$

$$(1.006)^n = \frac{7}{4}$$

$$\log 1.006^n = \log\frac{7}{4}$$

$$n = \frac{\log\frac{7}{4}}{\log 1.006}$$

$$n = 93.54\ldots$$

$\therefore$ it will take 94 months

Q 5 **2008**

Peter retires with a lump sum of \$100 000. The money is invested in a fund which pays interest each month at a rate of 6% per annum, and Peter receives a fixed monthly payment of $\$M$ from the fund. Thus, the amount left in the fund after the first monthly payment is $\$(100\,500 - M)$.

(i) Find a formula for the amount, $\$A_n$, left in the fund after n monthly payments.

(ii) Peter chooses the value of M so that there will be nothing left in the fund at the end of the 12th year (after 144 payments). Find the value of M.

Q9bi $P = 100000$, $R = 1 + \frac{6}{100\times 12} = 1.005$,

monthly payment $= M$.

$$A_n = 100000(1.005^n) - \frac{M(1.005^n - 1)}{0.005}$$

Q9bii $100000(1.005^{144}) - \frac{M(1.005^{144}-1)}{0.005} = 0$,

$$\therefore M = \frac{100000(1.005^{144})\times 0.005}{1.005^{144}-1} = 975.85$$

Q 6

2009

One year ago Daniel borrowed \$350 000 to buy a house. The interest rate was 9% per annum, compounded monthly. He agreed to repay the loan in 25 years with equal monthly repayments of \$2937.

(i) Calculate how much Daniel owed after his first monthly repayment.

(ii) Daniel has just made his 12th monthly repayment. He now owes \$346 095. The interest rate now decreases to 6% per annum, compounded monthly.

The amount, $\$A_n$, owing on the loan after the nth monthly repayment is now calculated using the formula

$$A_n = 346\,095 \times 1.005^n - 1.005^{n-1}M - \cdots - 1.005M - M$$

where $\$M$ is the monthly repayment, and $n = 1, 2, \ldots, 288$. (Do NOT prove this formula.)

Calculate the monthly repayment if the loan is to be repaid over the remaining 24 years (288 months).

(iii) Daniel chooses to keep his monthly repayments at \$2937. Use the formula in part (ii) to calculate how long it will take him to repay the \$346 095.

(iv) How much will Daniel save over the term of the loan by keeping his monthly repayments at \$2937, rather than reducing his repayments to the amount calculated in part (ii)?

Q8bi After one month and before repayment, amount owed $= 350000 \times \left(1 + \frac{\frac{9}{12}}{100}\right) = 352625$.

After repayment, amount owed $= 352625 - 2937 = \$349688$.

Q8bii $A_n = 0$ when $n = 288$.

$\therefore 346095 \times 1.005^{288} - \left(1 + 1.005 + 1.005^2 + \ldots + 1.005^{287}\right)M = 0$,

$346095 \times 1.005^{288} - \frac{1\left(1.005^{288} - 1\right)}{1.005 - 1}M = 0$,

$M = \$2270.31$

Q8biii $346095 \times 1.005^n - \frac{1\left(1.005^n - 1\right)}{1.005 - 1} \times 2937 = 0$,

$346095 \times 1.005^n - \frac{2937\left(1.005^n - 1\right)}{0.005} = 0$,

$346095 \times 1.005^n - 587400\left(1.005^n - 1\right) = 0$,

$1.005^n = 2.4342637$, $n = \frac{\ln 2.4342637}{\ln 1.005} \approx 178.37332$ months,

$\therefore$ it will take 14 years 11 months.

Q8biv Save $2270.31 \times 288 - 2937 \times 178.37332 = \$129967$ to the nearest dollar.

Q 7

2010

(i) When Chris started a new job, \$500 was deposited into his superannuation fund at the beginning of each month. The money was invested at 0.5% per month, compounded monthly.

Let $\$P$ be the value of the investment after 240 months, when Chris retires.

Show that $P = 232\,175.55$.

(ii) After retirement, Chris withdraws \$2000 from the account at the end of each month, without making any further deposits. The account continues to earn interest at 0.5% per month.

Let $\$A_n$ be the amount left in the account n months after Chris's retirement.

(1) Show that $A_n = (P - 400\,000) \times 1.005^n + 400\,000$.

(2) For how many months after retirement will there be money left in the account?

Q9ai The balance after n deposits is given by

$B_n = \dfrac{Q(R^n - 1)}{R - 1}$ where $Q = \$500$ and $R = 1.005$

.: $B_{240} = \dfrac{500(1.005^{240} - 1)}{0.005} = 231020.4476$

.: at the end of the 240[th] month,

$\$P = 231020.4476 \times 1.005 = \$232175.55$

Q9aii(1) $A_n = PR^n - \dfrac{Q(R^n - 1)}{R - 1}$ where $Q = \$2000$ and $R = 1.005$

.: $A_n = P \times 1.005^n - \dfrac{2000(1.005^n - 1)}{0.005}$

$= P \times 1.005^n - 400000(1.005^n - 1)$

$= (P - 400000) \times 1.005^n + 400000$

Q9aii(2) Let $A_n = (P - 400000) \times 1.005^n + 400000 = 0$

.: $1.005^n = \dfrac{-400000}{P - 400000} = \dfrac{-400000}{232175.55 - 400000} = 2.383443$

$n = \dfrac{\ln 2.383443}{\ln 1.005} \approx 174.14$, .: there will be money left in the account 174 months after retirement.

Q 8 — 2012

Ari takes out a loan of \$360 000. The loan is to be repaid in equal monthly repayments, \$$M$, at the end of each month, over 25 years (300 months). Reducible interest is charged at 6% per annum, calculated monthly.

Let \$$A_n$ be the amount owing after the nth repayment.

(i) Write down an expression for the amount owing after two months, \$$A_2$.

(ii) Show that the monthly repayment is approximately \$2319.50.

(iii) After how many months will the amount owing, \$$A_n$, become less than \$180 000?

Q15ci

$$\$A_1 = \$360000\left(1+\frac{6}{12\times 100}\right) - \$M = \$(360000\times 1.005 - M)$$

$$\therefore\ \$A_2 = \$(360000\times 1.005 - M)\times 1.005 - \$M$$

$$= \$(360000\times 1.005^2 - 1.005M - M)$$

or $\$A_2 = \$\left(360000\times 1.005^2 - \frac{M(1.005^2-1)}{1.005-1}\right)$ by formula.

Q15cii $A_{300} = 360000\times 1.005^{300} - \frac{M(1.005^{300}-1)}{1.005-1} = 0$

$$M = \frac{360000\times 1.005^{300}\times 0.005}{1.005^{300}-1} \approx 2319.485 \approx 2319.50 \text{ dollars}$$

Q15ciii $\$A_n < \$180000$,

$$360000\times 1.005^n - \frac{2319.485(1.005^n - 1)}{1.005-1} < 180000$$

$$360000\times 1.005^n - 463897.009(1.005^n - 1) < 180000$$

$$103897\times 1.005^n > 283897,\ 1.005^n > 2.732485$$

$n > \frac{\ln 2.732485}{\ln 1.005} = 201.54$, $\therefore$ after 202 months.

Q 9 — 2013

A family borrows \$500 000 to buy a house. The loan is to be repaid in equal monthly instalments. The interest, which is charged at 6% per annum, is reducible and calculated monthly. The amount owing after n months, \$$A_n$, is given by

$$A_n = Pr^n - M\left(1 + r + r^2 + \cdots + r^{n-1}\right), \qquad \text{(Do NOT prove this)}$$

where \$$P$ is the amount borrowed, $r = 1.005$ and \$$M$ is the monthly repayment.

(i) The loan is to be repaid over 30 years. Show that the monthly repayment is \$2998 to the nearest dollar.

(ii) Show that the balance owing after 20 years is \$270 000 to the nearest thousand dollars.

(iii) After 20 years the family borrows an extra amount, so that the family then owes a total of \$370 000. The monthly repayment remains \$2998, and the interest rate remains the same.
How long will it take to repay the \$370 000?

Q13di $n = 360$, $A_{360} = 0$, $P = 500000$, $r = 1.005$

In the formula, $1 + r + r^2 + \ldots + r^{n-1} = \frac{r^n - 1}{r-1}$.

$\therefore A_n = P.r^n - M\left(\frac{r^n-1}{r-1}\right)$, $0 = 500000\times 1.005^{360} - M\left(\frac{1.005^{360}-1}{0.005}\right)$

$\therefore\ M \approx 2997.75 \approx 2998$ dollars

Q13dii $n = 240$, $P = 500000$, $r = 1.005$, $M = 2998$

$$A_{240} = 500000\times 1.005^{240} - 2998\left(\frac{1.005^{240}-1}{0.005}\right)$$

$\approx 269903 \approx 270000$ dollars

Q13diii $P = 370000$, $r = 1.005$, $M = 2998$, $A_n = 0$

$$0 = 370000\times 1.005^n - 2998\left(\frac{1.005^n - 1}{0.005}\right).$$

Let $x = 1.005^n$, $370000x - 2998\left(\frac{x-1}{0.005}\right) = 0$, $\therefore\ x \approx 2.6115$

$1.005^n \approx 2.6115$, $n \approx 192.5$ months, $\therefore$ just over 16 years

Q10 **2014**

At the start of a month, Jo opens a bank account and makes a deposit of \$500. At the start of each subsequent month, Jo makes a deposit which is 1% more than the previous deposit.

At the end of every month, the bank pays interest of 0.3% (per month) on the balance of the account.

(i) Explain why the balance of the account at the end of the second month is

$$\$500(1.003)^2 + \$500(1.01)(1.003).$$

(ii) Find the balance of the account at the end of the 60th month, correct to the nearest dollar.

Question 16 (b) (i)

Let A_n be the balance at the end of nth month.

$$A_1 = 500(1.003)$$

$$A_2 = (A_1 + 500(1.01)) \times 1.003$$

$$= 500(1.003)^2 + 500(1.01)(1.003)$$

Question 16 (b) (ii)

$$A_2 = 500(1.003)^2 + 500(1.01)(1.003)$$

$$A_3 = 500(1.003)^3 + 500(1.01)(1.003)^2 + 500(1.01)^2(1.003)$$

...

$$A_n = 500(1.003)^n + 500(1.01)(1.003)^{n-1} + \ldots + 500(1.01)^{n-1}(1.003)$$

$$= 500(1.003)\left[(1.003)^{n-1} + (1.01)(1.003)^{n-2} + \ldots + (1.01)^{n-1}\right]$$

$$= 500(1.003)^n \times \frac{\frac{1.01}{1.003}^n - 1}{\frac{1.01}{1.003} - 1}$$

$$A_{60} = 44404$$

Q11 **2015**

Sam borrows \$100 000 to be repaid at a reducible interest rate of 0.6% per month. Let $\$A_n$ be the amount owing at the end of n months and $\$M$ be the monthly repayment.

(i) Show that $A_2 = 100\,000(1.006)^2 - M(1 + 1.006)$. **1**

(ii) Show that $A_n = 100\,000(1.006)^n - M\left(\frac{(1.006)^n - 1}{0.006}\right)$. **2**

(iii) Sam makes monthly repayments of \$780. Show that after making 120 monthly repayments the amount owing is \$68 500 to the nearest \$100. **1**

(iv) Immediately after making the 120th repayment, Sam makes a one-off payment, reducing the amount owing to \$48 500. The interest rate and monthly repayment remain unchanged. After how many more months will the amount owing be completely repaid? **3**

Question 14 (c) (i)

$$A_1 = 100000(1.006) - M$$

$$A_2 = A_1(1.006) - M$$

$$= 100000(1.006)^2 - M(1 + 1.006)$$

Question 14 (c) (ii)

$$A_3 = A_2(1.006) - M$$

$$= 100000(1.006)^3 - M(1 + 1.006 + 1.006^2)$$

$$\vdots$$

$$A_n = 100000(1.006)^n - M\underbrace{(1 + 1.006 + \ldots + 1.006^{n-1})}_{GP}$$

$$= 100000(1.006)^n - M\left(\frac{1.006^n - 1}{1.006 - 1}\right)$$

$$= 100000(1.006)^n - M\left(\frac{1.006^n - 1}{0.006}\right)$$

Question 14 (c) (iii)

$$M = 780.$$

$$A_{120} = 100000(1.006)^{120} - 780\left(\frac{1.006^{120} - 1}{0.006}\right)$$

$$= 68499.458\ldots \approx 68500 \text{ (nearest hundred)}$$

Question 14 (c) (iv)

The formula can be changed to: $A_n = 48500(1.006)^n - 780\left(\frac{1.006^n - 1}{0.006}\right)$.

Setting $A_n = 0$,

$$48500(1.006)^n = 780\left(\frac{1.006^n - 1}{0.006}\right)$$

$$48500(1.006)^n = 130000(1.006)^n - 130000$$

$$81500(1.006)^n = 130000$$

$$1.006^n = \frac{260}{163}$$

$$n = \frac{\ln\frac{260}{163}}{\ln 1.006} = 78.055\ldots$$

$\therefore$ After 79 months.

Q12 — 2016

A gardener develops an eco-friendly spray that will kill harmful insects on fruit trees without contaminating the fruit. A trial is to be conducted with 100 000 insects. The gardener expects the spray to kill 35% of the insects each day and that exactly 5000 new insects will be produced each day.

The number of insects expected at the end of the nth day of the trial is A_n.

(i) Show that $A_2 = 0.65(0.65 \times 100\,000 + 5000) + 5000$. **2**

(ii) Show that $A_n = 0.65^n \times 100\,000 + 5000\dfrac{(1-0.65^n)}{0.35}$. **1**

(iii) Find the expected insect population at the end of the fourteenth day, correct to the nearest 100. **1**

Q14bi $A_1 = 0.65 \times 100000 + 5000$

$$A_2 = 0.65A_1 + 5000 = 0.65(0.65 \times 100000 + 5000) + 5000$$

Q14bii $A_2 = 0.65^2 \times 100000 + 5000 + 0.65 \times 5000$

$$A_3 = 0.65A_2 + 5000 = 0.65^3 \times 100000 + 5000 + 0.65 \times 5000 + 0.65^2 \times 5000$$

$$A_n = 0.65^n \times 100000 + S_n = 0.65^n \times 100000 + \frac{5000(1-0.65^n)}{0.35}$$

Q14biii $A_{14} = 0.65^{14} \times 100000 + \dfrac{5000(1-0.65^{14})}{0.35} \approx 14500$

Q13 — 2017

Anita opens a savings account. At the start of each month she deposits $X into the savings account. At the end of each month, after interest is added into the savings account, the bank withdraws $2500 from the savings account as a loan repayment. Let M_n be the amount in the savings account after the n^{th} withdrawal.

The savings account pays interest of 4.2% per annum compounded monthly.

(i) Show that after the second withdrawal the amount in the savings account is given by **2**

$$M_2 = X(1.0035^2 + 1.0035) - 2500(1.0035 + 1).$$

(ii) Find the value of X so that the amount in the savings account is $80 000 after the last withdrawal of the fourth year. **3**

Question 15 (b) (i)

Sample answer:

$$M_1 = X \times \left(1 + \frac{0.042}{12}\right) - 2500$$

$$= X(1.0035) - 2500$$

$$M_2 = ((X(1.0035) - 2500) + X)1.0035 - 2500$$

$$= X(1.0035)^2 + X(1.0035) - 2500(1.0035) - 2500$$

$$= X(1.0035^2 + 1.0035) - 2500(1.0035 + 1)$$

Question 15 (b) (ii)

Sample answer:

At the end of 4 years = 48 months, $M_{48} = 80\,000$

$$X(1.0035^{48} + \cdots + 1.0035) - 2500(1.0035^{47} + \cdots + 1) = 80\,000$$

$$X\left[\frac{1.0035(1.0035^{48}-1)}{0.0035}\right] - \frac{2500(1.0035^{48}-1)}{0.0035} = 80\,000$$

$$X\left[\frac{1.0035(1.0035^{48}-1)}{0.0035}\right] = 210\,421.2054$$

$$52.351X = 240\,421.2054$$

$$X = 4019.42$$